OS Practical Exam Question Bank

1. Write a C program to implement FCFS (with arrival time=0 for all) Calculate waiting time, turnaround time for each process. Calculate avg. waiting time, avg turnaround time for process.
2. Write a C program to simulate Non Pre-emptive Priority Scheduling policy with arrival time
3. Write a C program to simulate the Preemptive SJF scheduling policy.
4. Write a program to implement non-prememptive SJF (with arrival time=0 for all) Calculate waiting time, turnaround time for each process. Calculate avg. waiting time, avg turnaround time.
5. Write a program to implement Round Robin. Calculate waiting time, turnaround time for each process. Calculate avg. waiting time, avg turnaround time.
6. Write a C program to simulate the Best Fit Memory Allocation Technique.
7. Write a C program to simulate the Worst Fit Memory Allocation Technique.
8. Write a program to simulate the First Fit Memory Allocation Technique.
9. Write a C program to simulate FIFO Page replacement.
10. Write C Program to simulate LRU Page Replacement.
11. Write C Program to simulate Optimal Page Replacement.
12. Write C Program to simulate Paging technique.
13. Write a program to implement the FCFS Disk Scheduling Policy
14. Write C Programto implement the SSTF Disk Scheduling Policy
15. Write C Program to implement the following SCAN Disk Scheduling Policy
16. Write C Program to implement the following C-SCAN Disk Scheduling Policy.
17. Write C Program to implement the following LOOK Disk Scheduling Policy.
18. Write C Program to implement the following C-LOOK Disk Scheduling Policy
19. Write a C program to implement solution of Producer consumer problem through Semaphore.
20. Write a program to demonstrate the concept of deadlock avoidance through Banker's Algorithm.

21. **Any problem statement on Shell Programming and Commands practiced in first experiment 1 and 2 can come in OS practical exam. Practice questions are given as follows for same:**

Practice Problem Statements on shell programming and commands

1) Write a command to create two input files containing five names each and perform the following operations on the command prompt:

a. Concatenate two files.

b. Arrange the names of the friends in ascending order.

c. Arrange the names of the friends in descending order.

d. Remove the contents of the files.

e. Repeat the entire work via a shell program

2) Write a shell program to determine whether the entered 3-digit no is Armstrong or not

3)Write a shell program to compute the Fibonacci series using a while loop.

4) Write a shell program to Determine whether a person is eligible to vote.

5) Write a shell program to do the following:

1. Display the top 10 processes in descending order.
2. Display the process with the highest memory usage.
3. Display the current user logged in and login name.
4. Display the first 11 lines from a file.

6) Write a shell program to create two employee files, and each should store the following information: 1) Empid 2) Empname 3) Salary 4) Designation and perform the following operations: a) Join two files in 3rd file and display two files (concatenate two files). b) Count the contents of the file.

7) Write a shell program to create two customer files, and each should store the following information: 1) Custid 2) Custname 3) City and 4) Balance and perform the following operations: a) Join two files in 3rd file and display two files (concatenate two files). b) Count the contents of the file.

8) Create a directory named OS_Practical. Inside this directory, create three files: file1.txt ,file2.txt and file3.txt. Copy file1.txt to a new file named file1_backup.txt. Move file2.txt to the Backup directory. Rename file3.txt to updated_file3.txt using the mv command. Remove the entire Backup directory along with its contents.

9) Write a shell program to find sum of digits of entered no.

10) Write a shell program to swap 2 numbers.

11) Write a shell program to accept Percentage marks (integer) and display the grade.

12) Write a shell program to print the multiplication table of number n.

13) Write a shell program to check whether the no is even or odd.

14) Write a shell program to determine if a person is eligible to vote or not

15) Write a shell program to find sum of n numbers.

16) Write a shell program to determine if a person is eligible to vote or not.

17) Write a shell program to find out the factorial of given no.

18) Write a shell program to find sum of 1st n odd numbers.